MSML606 - Project 2 Proposal

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1. Problem Statement

Every day, Amazon deploys thousands of delivery drivers across the country, each responsible for dropping off packages at dozens of addresses within a defined area. The question each route planner must answer is deceptively simple: in what order should a driver visit a set of delivery stops so that the total distance traveled is minimized? On the surface this sounds like a scheduling problem, but it belongs to a deeper and far more difficult class of combinatorial optimization problems. With just 10 stops, there are over three million possible orderings. With 15 stops, that number exceeds 87 billion. No human planner, and no naive algorithm, can evaluate all of them in reasonable time.

This is a direct instance of the Travelling Salesman Problem (TSP), one of the most well-studied problems in computer science and a canonical example of NP-hard combinatorial optimization. The goal of my project is to model a realistic Amazon last-mile delivery route as a TSP instance and solve it using the Held-Karp dynamic programming algorithm, demonstrating how algorithmic thinking can dramatically reduce the search space and produce exact or near-optimal solutions that are far beyond the reach of intuitive approaches.

2 Algorithms that I plan to explore

In this work, I will focus on solving the problem using dynamic programming (DP). Classic DP applications include Longest Common Subsequence (LCS), the 0/1 Knapsack problem, and the Travelling Salesman Problem. My project focuses specifically on TSP, applying the Held-Karp algorithm as the primary DP solution.

2.1 Held-Karp

Held-Karp reformulates the TSP by defining a subproblem: what is the minimum cost to travel from the depot, visit exactly a given subset of stops, and arrive at a specific stop within that subset? It solves this question for all subsets of increasing size, building up from two-node base cases to the full route. Because each larger subproblem reuses previously computed smaller answers, the algorithm runs in $O(2^n \times n^2)$ time rather than the $O(n!)$ required by brute force. For 15 delivery stops, this reduces the number of computations from over 87 billion to roughly 7,000.

For larger route sizes where exact DP becomes computationally expensive, I will also implement a 2-opt heuristic as a scalable complement. This allows my application to demonstrate both the exact solution (Held-Karp, up to 15 nodes) and a high-quality approximation (2-opt, for larger inputs), with a side-by-side comparison against a greedy nearest-neighbor baseline.

3. Dataset

I will use the Amazon Last Mile Routing Research Challenge dataset ([HuggingFace link]()). This is a publicly available dataset released by Amazon Science in collaboration with MIT. The dataset contains real-world routing data from Amazon delivery operations across five major US cities, including actual stop sequences, geographic coordinates, package attributes, and driver-executed route outcomes. It was originally published as part of a machine learning challenge aimed at learning high-quality routing policies from historical driver behavior.

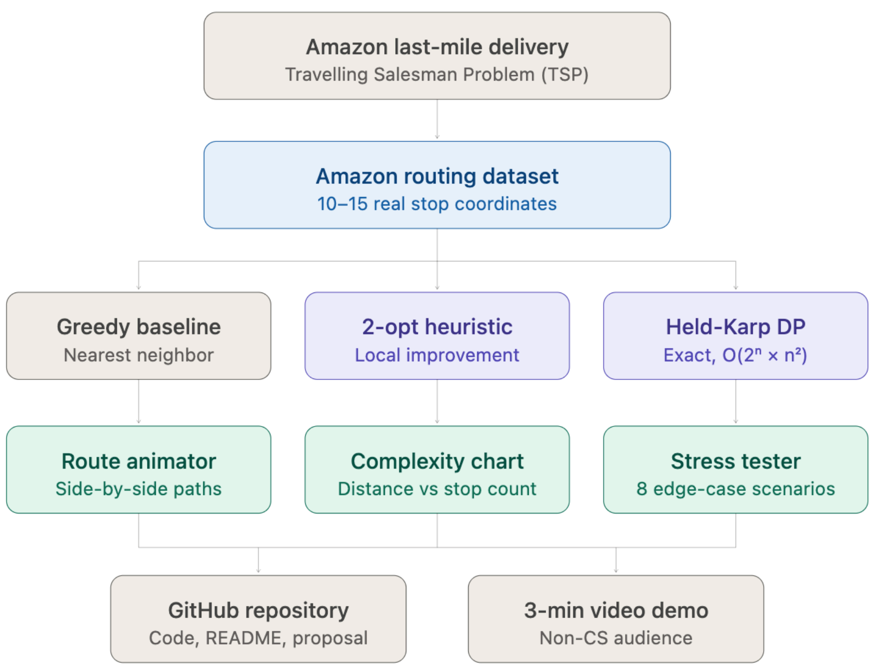
For my demo, I will extract a subset of 10 to 15 delivery stop coordinates from a single route instance within the dataset, representing a realistic neighborhood-level delivery scenario. This subset is small enough for Held-Karp to solve exactly, yet large enough to make the efficiency gain over naive approaches visually compelling.

4. Final Application and Demo

The final application will be an interactive visual demo designed for a non-technical audience. It will present a map of delivery stops drawn from the dataset and animate three routing strategies side by side: a random greedy route, a nearest-neighbor greedy route, and the optimal Held-Karp DP route. Each strategy will display its total distance and step count in real time as the path is drawn, making the efficiency difference immediately visible without requiring any background in algorithms.

The demo will also include a complexity simulation that plots total route distance against increasing numbers of delivery stops, showing how the greedy approach degrades faster than the optimized solution as the problem grows. A scenario stress-tester will allow viewers to select geometric edge cases such as clustered stops, ring layouts, and adversarial configurations, demonstrating conditions where greedy routing fails most dramatically.

The narrative framing throughout the demo will remain grounded in the delivery context: every unnecessary mile driven represents wasted fuel, increased emissions, and slower service. The goal is for anyone watching the three-minute video to walk away understanding not just that the DP solution is better, but intuitively why it is better, and why that matters in practice.



5. AI usage

- For implementing the code for Held-Karp DP and 2-opt heuristic, I have taken reference of an existing tutorial blog (link)
- I am using Perplexity AI to help me with the UI for the code.